

Fahim Shahryer

AI/ML Engineer

BSc in Computer Science & Engineering

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🌐 <https://github.com/fahimshahryer>

🌐 <https://fahimshahryer.github.io/>

Professional Summary

AI/ML Engineer(MLOps) with 2+ years of experience designing and deploying production-grade LLM systems, agentic architectures, and Retrieval-Augmented Generation (RAG) pipelines. Experienced in multi-LLM orchestration, multi-tenant SaaS AI platforms, and scalable backend systems using FastAPI and AWS. Built and deployed AI solutions used by 100+ researchers and enterprise teams, focusing on robust system design, cost-aware inference, and real-world deployment rather than experimental prototypes. Strong foundation in first-principles problem solving, system architecture, and end-to-end ML lifecycle management.

Professional Experience

MLOps Engineer

Oct 2025 - Present

Cloudscockpit

- Architected and deployed **Actionboard AI**, a multi-tenant AI-powered project management SaaS platform built around conversational **AI agents**, enabling structured task planning, document analysis, and role-based guidance through natural chat interactions
- Designed a production-grade **RAG pipeline** with global and tenant-level knowledge bases — implemented document ingestion (PDF, DOCX) via AWS S3, chunking, embedding, and semantic retrieval using **PostgreSQL (pgvector)** for grounded and context-aware responses
- Integrated multiple **LLM providers** (OpenAI, Anthropic, Gemini, Grok) and implemented a custom multi-step reasoning framework (RAOARA) for structured task decomposition and agent orchestration
- Built scalable multi-tenant infrastructure with **FastAPI**, **LangGraph**, **Neo4j**, **Redis**, and AWS services (ECS, EC2, RDS, Cognito, ElastiCache), including real-time streaming responses, MCP tool servers, and tenant-level token usage analytics

AI/ML Engineer

April 2025 - Oct 2025

Periscope Labs, Dhaka

- Led **Agentic AI** framework for document validation using **RAG** architecture and multi-**LLM** approach (OpenAI, Mistral AI) to automatically validate invoices and policy documents against business rules, reducing manual review by 70%
- Engineered an **admin dashboard** for policy document management and control with **annotation** pipeline that identifies and marks policy violations directly in documents
- Designed loan default prediction pipeline using **time series analysis** and feature engineering, creating an **early warning system** that identifies at-risk customers 30 days in advance with 75% prediction accuracy

Research Assistant

Jan 2024 - May 2025

Quantum Robotics & Automation Research Group, MIST

- Engineered an end-to-end **AutoML** solution (AKI) assisting 100+ researchers with intuitive flow covering data upload, preprocessing, visualization, model training, and deployment, eliminating coding barriers
- Built computer vision and AI-based assistive interface achieving 92% accuracy in **eye movement tracking**, empowering paralyzed and post-surgery patients with voice- and hand-free communication capabilities
- Directed 3 cutting-edge AI research projects contributing to research publications while mentoring 7+ junior researchers in machine learning projects

EDGE-AI Trainee

Aug 2023 - Nov 2023

ITIL - World Bank Funded Program

- Completed intensive 160-hour AI specialization program funded by World Bank in partnership with Bangladesh Computer Council, focusing on enterprise-level AI implementation strategies
- Mastered industry best practices in AI deployment, model optimization, and production workflows, achieving top 5% performance in program assessments and practical projects

Education

BSc in Computer Science and Engineering

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

Feb 2019 - Dec 2023

GPA: 3.18/4.00

- Thesis: Development of an intelligent system for eye motion tracking and action prediction

Key Projects

Automated Document Processor

Demo

Tech Stack: Python, LangChain, LLMs, RAG, Mistral, Agentic AI, Weaviate, Docker

NexNews - AI News Fact Checker

Demo

Tech Stack: Python, LLMs, AI Agents, News API, Web Scraping

AutoML - Artificial Kinetic Intelligence (AKI)

Demo

Tech Stack: Scikit-Learn, ML Algorithms, Plotly, Python, MongoDB

Eye Motion Tracking and Action Prediction Assistive System

Demo

Tech Stack: Scikit-Learn, Machine Learning, TensorFlow, Computer Vision, Python

Controlling Robotic Hand from Remote Distance

Demo

Tech Stack: Computer Vision, Arduino, Robotics

Child Companion AI Robot

Demo

Tech Stack: Computer Vision, Neural Network, Mobile App(Flutter), IoT, Raspberry Pi, Arduino,

Technical Expertise

AI and LLM Systems: Agentic AI, RAG Pipelines, Multi-LLM Orchestration, Prompt Engineering, Fine-Tuning, Embedding Pipelines, Vector Search, Multi-Tenant AI Architecture

Frameworks and Libraries: LangChain, LangGraph, LlamaIndex, FastAPI, PyTorch, TensorFlow, Scikit-Learn, OpenCV

Backend and APIs: Python, REST APIs, Real-Time Streaming, LLM APIs (OpenAI, Anthropic, Gemini, Grok)

Databases: PostgreSQL (pgvector), Neo4j (Graph DB), Redis, MongoDB

Cloud and DevOps: AWS (ECS, EC2, RDS, S3, Cognito, ElastiCache), Docker, GitHub Actions (CI/CD), Model Deployment, Infrastructure Design

Programming: Python, C/C++, SQL

Soft Skills: System Design Thinking, Research, Leadership, Cross-Functional Collaboration

Awards & Recognition

Champion - AI Hackathon 2025 (Healthcare)

Certificate

Bangladesh Innovation Conclave — Managed a team of 3, designed an AI application for healthcare sector

Highest-Ranked Submission - BEAR Summit 2025 (AI Onsite Display)

Certificate

Code AI-048 — Qualified for on-site display

1st Runner-Up - BEAR Summit 2025 (AI Demo)

Certificate

Bangladesh National Semiconductor Symposium — submission code AI-047

Global Champion - University Rover Challenge

Certificate

University Rover Challenge - The Mars Society — Competed against 20+ teams worldwide

21st Century Employability Skilling Program - Advanced

Certificate

Wadhvani Foundation — Advanced professional development certification

EDGE-AI Certification

Certificate

Top 5% performance — Internship

5th Place - Technovent 2023 Hackathon

Certificate

SUST — 48-hour AI/ML innovation challenge